Time Series Analysis: Session Learning Objectives

After this session, you will be able to

1. Explain the form of two typical time series models:
   - Additive model
   - Multiplicative model
2. Estimate the Trend/ Seasonal/ Cyclical effects in a series
3. Describe the procedure for extracting these effects from a series
4. Perform the decomposition for a given series manually and on R
5. Describe the use of such extracted patterns of a series

Recall Classical Decomposition

- Components of a Time Series:
  - Trend(T_t): Long term movement in the mean
  - Seasonal variation(S_t): Short-term fluctuations, usually assumed to be within a year henceforth in these notes
  - Cycle(C_t): long-term cyclical patterns e.g. sun-spots, business cycles[2]
  - Residuals (R_t): random/other unexplained variation components
- For seasonality, need to decide whether the series is better represented by a:
  - Additive model, i.e. $y_t = T_t + S_t + R_t$
  - Multiplicative model, i.e. $y_t = T_t \times S_t \times R_t$

[2] These are not considered in detail on the course

Example 2.3: A Simple Example

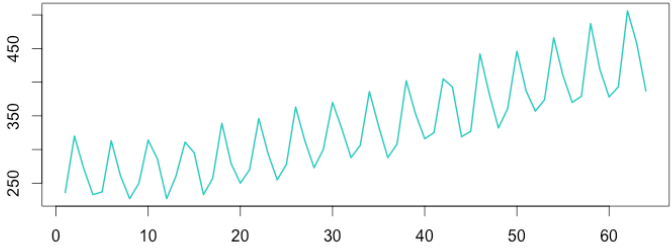


FIGURE 2.21: Beer Consumption

- Fig 2.21 shows a typical real life time series.
- As will be seen, little variation in amplitude with time.
- As will be seen, series like this would suit the use of an additive model.

Example 2.4: Another Simple Example

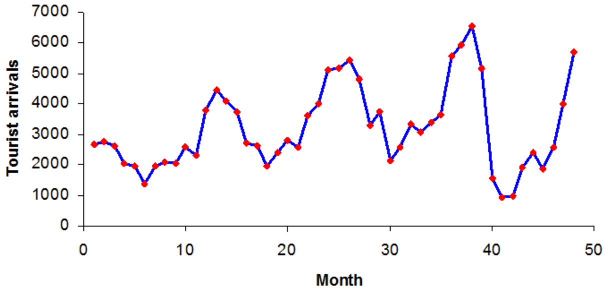


FIGURE 2.22: Sri Lankan Tourist Arrivals

- Fig 2.22 shows a typical real life time series.
- Notice the slight increase in variation with increasing time.
- As will be seen, series like this require using a multiplicative model.

For Decomposition - Which Model Suits?

Additive or Multiplicative Model?

- Additive model
  - Suits if the *size* of seasonal fluctuations (or variation around the trend T_t) doesn't vary with the time series level[3].
  - Means the seasonality is the same (roughly constant) in same period over different years (does not depend on level)
- Multiplicative model
  - Suits if the variation in seasonal pattern or that around the trend (T_t) *does* vary with the time series level.
  - With economic time series, such models are common.
  - Sometimes seasonal effect is a proportion of underlying trend value, e.g. in previous slide, they increase with trend

[3] Level can roughly be described as long-term mean of a time series

Example 2.5: A More Complex Example

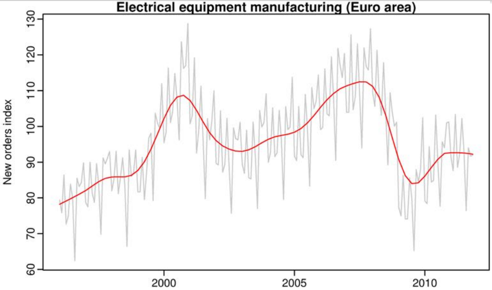


FIGURE 2.23: Some Electrical Equipment Order Data

- Figure 2.23 shows a graph of Electrical Equipment Orders.
- The trend component is shown in red, raw data in grey.
- While the raw data oscillates greatly, the size of variation around the level is roughly constant
- Next slide shows an additive decomposition of this data.

Example 2.5: A More Complex Example (/2)

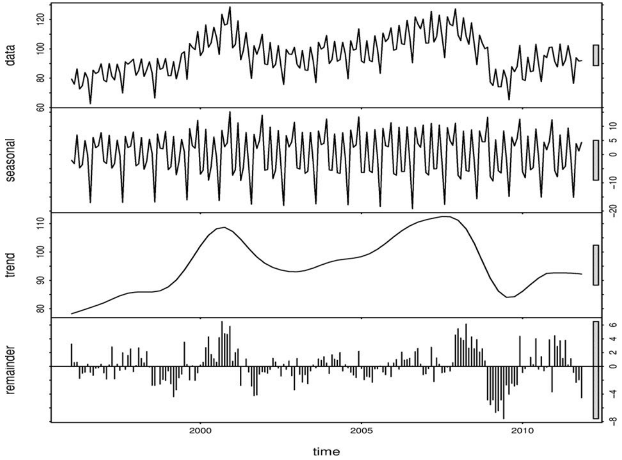


FIGURE 2.24: Additive Decomposition of Figure 2.23

Example 2.5: A More Complex Example (/3)

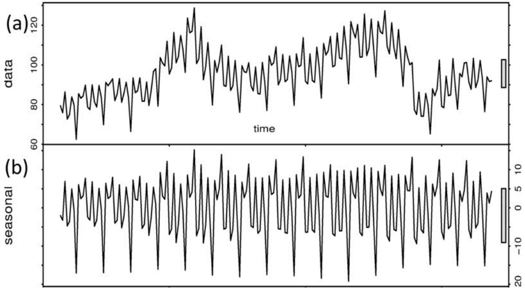


FIGURE 2.25: Original Data & Seasonal from Figure 2.24

- The top part of Fig 2.24 enlarged in Fig 2.25
- Data in (b), (c) & (d) (next slide) sum to original data in (a).
- Note (b) varies slowly over time, so any 2 years in a row have similar patterns, but years far apart may have different ones.

Example 2.5: A More Complex Example (/4)

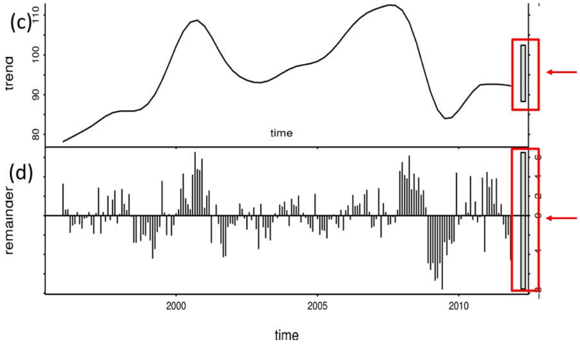


FIGURE 2.26: Trend & Remainder from Figure 2.24

- (d) is remainder when seasonality, trend taken from the data.
- The bars at right show the relative scales of components.
- Lengths are same but sizes vary as plots are on different scales.

Aside: Estimating the Seasonal Effect

- Recall that seasonal effects occur when the series repeats systematically in short time periods (often within a year)
- A de-seasonalised series shows the pattern of change over time with all seasonal effects removed.
  - allows direct comparisons between time points in this series, unaffected by seasonal changes
- First de-trend the series by finding either:

$D_i = Y_i - T_i$ additive model or $D_i = Y_i / T_i$ multiplicative model

- Means of detrended values D_i are scaled so seasonal mean:
  - averages to zero for an additive model, or
  - averages to 1 for a multiplicative model.
- Seasonal means are often called *Seasonal Indices (SI)* and expressed as a percentage value.

Seasonally–Adjusted (SA) Data

- If seasonality is not the main focus, SA data can be of use.
- Example: Monthly Unemployment Data
  - *Seasonal Variation*: Jobless rises from job-seeking school leavers
  - *Non-Seasonal Variation*: Jobless rises due to large employers laying off workers.
  - Most studying jobless data focus on non-seasonal variation.
  - Usually SA to show variation due to underlying state of the economy rather than seasonal variation.
  - So employment (and other economic) data are usually SA.
- SA series contain remainder & trend components.
- So they not 'smooth' & 'downturns'/'upturns' can mislead.
- If turning points in series are the focus & must interpret series changes, better to use trend not SA data.

Example 2.5: A More Complex Example (/5)

- If the seasonal part is removed from the raw data, resulting data are Seasonally Adjusted.
- Fig 2.27 shows seasonally adjusted electrical equipment orders

FIGURE 2.27: Some Electrical Equipment Order Data

Example 2.6: Births Data 1982-92

Year\Quarter	Q1	Q2	Q3	Q4	Totals
1982	17474	18475	18258	16726	70933
1983	17087	17260	17024	15444	66815
1984	16424	16218	16155	15440	64237
1985	15913	16012	15683	14642	62250
1986	15094	16663	15164	14504	61425
1987	14961	15406	14967	13530	58864
1988	13967	14426	13857	12050	54300
1989	13133	13947	13158	11421	51659
1990	13321	13989	13285	12359	52954
1991	13438	13376	13620	12256	52690
1992	13565	13399	13150	11470	51584

TABLE 2.6: Quarterly Irish Registered Births data (1982 to 92)

- Data seems to change wildly between quarters - expected?
- Totals column (right) shows trend seems to be downwards.

Example 2.6: Births Data 1982-92 (/2)

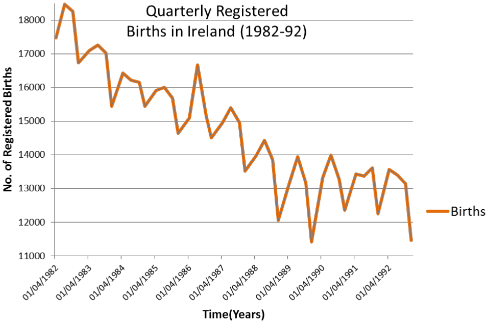


FIGURE 2.28: Plot of Table 2.6 Births Data

- Shows an annual seasonality – not obvious in raw data.
- Annual Q4 data (Oct-Dec) lower than other 3 quarters.
- Fluctuations & general trends commented on are obvious.

Ex 2.6 Births 1982-92 (/2): Staged Calculations

No of Births Reg'd 01/01/82-31/03/82

4pt MA value at 01/07/82

2X4pt MA Trend Value for Q3 15/08/82

Dates (Qtr end)	Births	4pt MA (nb mid-Qtr!)	Trend (2X4 pt MA)
01/04/1982	17474		
01/07/1982	18475		
01/10/1982	18258	17733.25	17684.875
31/12/1982	16726	17636.5	17484.625
01/04/1983	17087	17332.75	17178.5
01/07/1983	17260	17024.25	16864
01/10/1983	17024	16703.75	16620.875
31/12/1983	15444	16538	16407.75
01/04/1984	16424	16277.5	16168.875
01/07/1984	16218	16060.25	16059.75
01/10/1984	16155	16059.25	15995.375
31/12/1984	15440	15931.5	15905.75
01/04/1985	15913	15880	15821
01/07/1985	16012	15762	15662.25
01/10/1985	15683	15562.5	15460.125
31/12/1985	14642	15357.75	15439.125
01/04/1986	15094	15520.5	15455.625
01/07/1986	16663	15390.75	15373.5
01/10/1986	15164	15356.25	15339.625
31/12/1986	14504	15323	15165.875
01/04/1987	14961	15008.75	14984.125

Ex 2.6 Births 1982-92 (/2): Staged Calculations

Dates (Qtr end)	Births	4pt MA (nb mid-Qtr!)	Trend (2X4 pt MA)	Detrended Values D(i)=Births(i)/ Trend(i)	Qly Mean of D(i)= Seasonal Indeces (SI)	Seasonally Adjusted SA(i)=Births(i)/SI(i)
01/04/1982	17474					
01/07/1982	18475					
01/10/1982	18258	17733.25	17684.875	1.032407636	1.01761154	17942.0136
31/12/1982	16726	17636.5	17484.625	0.95661188	0.93677457	17854.8826
01/04/1983	17087	17332.75	17178.5	0.994673575	1.005952854	16985.8855
01/07/1983	17260	17024.25	16864	1.023481973	1.041137547	16578.0209
01/10/1983	17024	16703.75	16620.875	1.024254138	1.017611536	16729.3701
31/12/1983	15444	16538	16407.75	0.941262513	0.936774572	16486.3570
01/04/1984	16424	16277.5	16168.875	1.015778772	1.005952854	16326.8089
01/07/1984	16218	16060.25	16059.75	1.009853827	1.041137547	15577.1925
01/10/1984	16155	16059.25	15995.375	1.009979447	1.017611536	15875.4097
31/12/1984	15440	15931.5	15905.75	0.970718137	0.936774572	16482.0870
01/04/1985	15913	15880	15821	1.005815056	1.005952854	15818.8328
01/07/1985	16012	15762	15662.25	1.022330763	1.041137547	15379.3320
01/10/1985	15683	15562.5	15460.125	1.014416119	1.017611536	15411.5784
31/12/1985	14642	15357.75	15439.125	0.948369807	0.936774572	15630.2278
01/04/1986	15094	15520.5	15455.625	0.97660237	1.005952854	15004.6793
01/07/1986	16663	15390.75	15373.5	1.083878102	1.041137547	16004.6096
01/10/1986	15164	15356.25	15339.625	0.988550894	1.017611536	14901.5606
31/12/1986	14504	15323	15165.875	0.956357612	0.936774572	15482.9139
01/04/1987	14961	15008.75	14984.125	0.9984567	1.005952854	14872.4664

No of Births Reg'd 01/01/82-31/03/82

4pt MA value at 01/07/82

2X4pt MA Trend Value for Q3 15/08/82

Detrended Value at 15/02/83= 17087/17178.5

SI Value at 15/08/85= ∑ Values (some not shown)/ no. Seasonal Values

Seasonally-Adj Value at 15/08/85 = Births Value/ Average Seasonal = 15683/1.0176

- Detrended values are Birth values/these Trend values
- Seasonal Indices (SI) are means of quarterlies, D_i (not all shown)
- SA or deseasonalised numbers are Births/SI values.

Ex 2.6 Births 1982-92 (/4): Trend Component

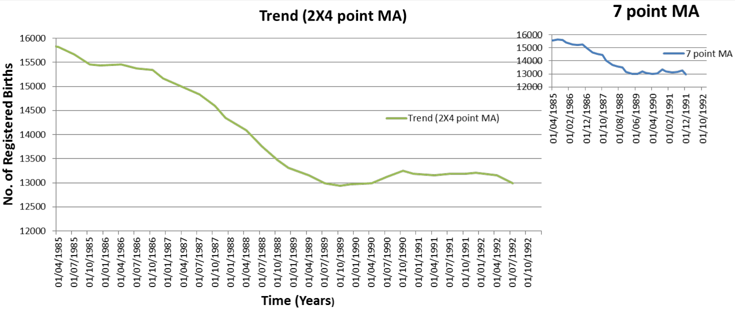


FIGURE 2.29: Irish Registered Births data Trend (1985 to 92)

- Compare main to that calculated using a 7 point MA (inset)
- Main is smoother than 7 pt MA which is based on an equal weighting of 7 data points rather than a WMA.

Ex 2.6 Births 1982-92 (/5): Seasonal Component

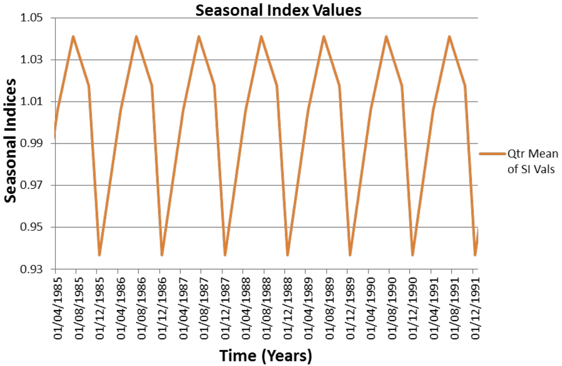


FIGURE 2.30: Irish Registered Births Data Seasonality (1985 to 92)

- Seasonal Indices shown represent raw data's seasonal part
- Large seasonal variation is obvious on the scale shown.

Ex 2.6 Births (/6): Raw & Seasonally Adjusted

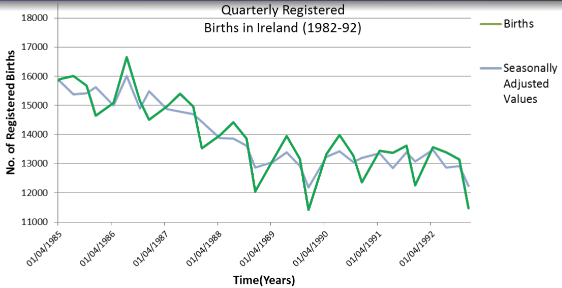
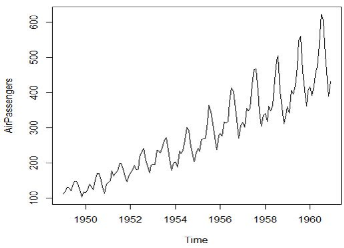


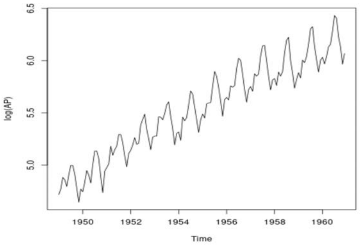
FIGURE 2.31: Raw & Seasonally Adjusted (SA) Births Data (1985 to 92)

- Shows SA smoother than raw, rougher than trend series.
  - SA more important than raw due to its more scientific basis – not as much subjectivity as trend estimation
  - SA level from 1989 – except for ‘89 Q4, ‘92 Q4 (maybe social trends or macroeconomic circumstances)

Example 2.7: Australian Air Passengers Data



(a) Raw Air Pax Data



(b) Log-Transformed Air Pax

FIGURE 2.32: Monthly International passengers (in Millions) 1949-60

- Air Pax data in Fig. 2.32(a) shows significant variation with level.
  - As a result, would probably suit a Multiplicative Decomposition.
  - This is given by: $y_t = T_t \times S_t \times R_t$
  - Can treat using logs Fig. 2.32(b) and an Additive model: $log(y_t) = log(T_t) + log(S_t) + log(R_t)$

Ex 2.7 Air Passengers (/2): Full Decomposition

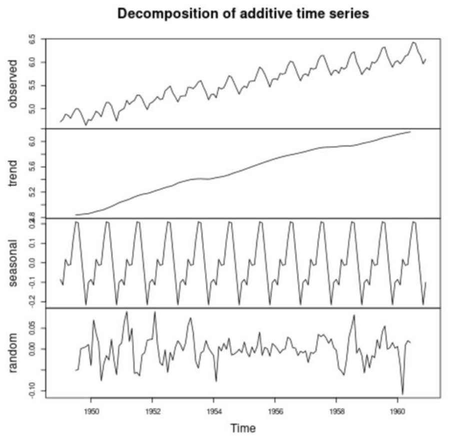


FIGURE 2.33: Log-Tranformed Air Passengers: Full Decomposition

- Full decomposition of log transformed series is shown in Fig 2.33
- Patterns are clear: especially trend & seasonal

Lecture Summary

In this lecture we have looked at different models that we can use to decompose time series data:

- An Additive Model:
  - Model has the form of a sum of trend, seasonal & residual components
  - Suits when seasonal variation doesn't vary with level
- A Multiplicative Model:
  - Model has the form of a product of trend, seasonal & residual components
  - Suits when seasonal variation varies with level
- Such decomposition can be done using moving averages or with a package like *R*